Sarthak Mistry

sarthakmistry44@gmail.com — +1 (930) 904-4600 — linkedin.com/in/sarthak-mistry — github.com/Sarthakmistry — Open to relocation

SUMMARY

Data Engineer with 2.5+ years building production batch and real-time pipelines (Databricks, Kafka, Debezium, Airflow, dbt, Snowflake) at scale: 5TB+ ingested daily, 2PB structured and unstructured data managed, ~30% cloud cost cut. Skilled in ETL, dimensional data modeling, data quality (Great Expectations, dbt tests), data governance, and AWS.

PROFESSIONAL EXPERIENCE

Indiana University, Dept. of Statistics ***Jan 2026 – Present***

Research Assistant — Data Infrastructure

- Developing a CGM cyber-attack detector, reaching **0.998 AUC-PR** in development testing, by building a leakage-free time-series pipeline with feature engineering and BART model.
- Enabled portable, reproducible Bayesian ML workflows across clouds by adding AWS S3, GCS, and Azure Blob backends and a Dockerized RStudio environment to the OpenBT project.

Piramal Capital & Housing Finance ***Jul 2022 – Jul 2024***

Data Engineer

- Built batch ETL pipelines using Databricks, ingesting **5TB+** of data dailty from Salesforce, MongoDB, SQL Server, and REST APIs into Delta Lake and Snowflake, cutting loan-approval turnaround by ~25%.
- Deployed real-time CDC pipeline (Debezium, Kafka, Apache Flink) processing **450K+ updates/hour**, enabling live monitoring of **$500M** in monthly loan disbursements on Apache Superset.
- Implemented a 150-feature regulatory data mart using dbt models on Snowflake, with tested, version-controlled transformations, cutting the compliance team's audit-reporting turnaround by ~40%.
- Designed **PII-masked** feature pipelines (Databricks, Delta Lake) combining S3 and Snowflake data for credit-risk models on SageMaker, achieving **93% masking precision** across **7M+** customer records.
- Reduced monthly **cloud spend by ~30%** through Databricks job clusters with spot instances and Snowflake warehouse right-sizing, while maintaining daily refresh **SLAs across ~150TB warehouse**.
- Partnered with the Chief Economist to automate an economic-indicator scraping pipeline (Python, Selenium, Airflow) into Snowflake, reducing manual effort by 73% and enabling weekly macro reporting.

NSEIT – Dex Analytics ***Oct 2021 – Mar 2022***

Data Engineering Intern

- Scaled post-exam analytics for a **7.3M** candidate recruitment drive by building Spark batch pipelines on PostgreSQL/SQL Server, cutting scorecard query latency by 40%.
- Designed dbt schema validations and data quality tests on PostgreSQL for proctoring data from **237+ exam centers**, reducing ingestion errors by 25% for Kibana dashboards.

PORTFOLIO PROJECTS

E-commerce Data Pipeline — *Airflow, dbt, Snowflake, Great Expectations* GitHub

- Built production-grade medallion pipeline (RAW -> staging -> Data Vault -> star schema/OBT) across 17 dbt models on TPC-DS SF10 (~10GB, 25 tables) in Snowflake.
- Orchestrated Airflow DAGs (Astro CLI, Astronomer Cosmos) with per-model retries; enforced 50+ Great Expectations checks across 7 suites with quarantine routing for analyst triage instead of hard pipeline failure.
- Modeled Data Vault 2.0 with MD5 hash-diff and SCD Type 2 history; built OBT with RFM customer segmentation.

Real-Time Transaction Streaming & Anomaly Detection — *Kafka, Spark, Hudi, Trino* GitHub

- Designed self-managed 3-node cluster (Kafka, Spark, Trino) processing ~500 live Bitcoin transactions/min from a WebSocket feed.
- Deployed SparkML KMeans anomaly-detection model as a Spark Structured Streaming job scoring transactions in-flight, persisting results to Apache Hudi tables on HDFS.
- Exposed scored results through Trino with Hive Metastore for low-latency ad-hoc SQL investigation of flagged transactions.

Snowflake Lineage — *React, Vis.js, Django, Snowflake* GitHub

- Built full-stack lineage tool (React/Vis.js, Django API) mapping Snowflake table and column dependencies from account metadata to increase observability.
- Cut column-level impact analysis from ~30 minutes of manual SQL tracing to under 10 seconds of visual exploration.

TECHNICAL SKILLS

Languages: Python, SQL, Bash, JavaScript
Data Engineering: Snowflake, dbt, Apache Airflow, Kafka, Spark (PySpark, Structured Streaming), Debezium CDC, Trino/Presto, Apache Hudi, ELT/ETL
Data Modeling & Quality: Star Schema, Data Vault, SCD Type 2, Kimball Dimensional Modeling, OBT, Great Expectations
Cloud & Infrastructure: AWS (S3, Lambda, EMR), GCP (BigQuery), Docker, Git, CI/CD
Governance & BI: Snowflake Dynamic Data Masking, RBAC, PII Detection, Grafana, Power BI, Streamlit

EDUCATION

University of Mumbai **Aug 2018 – May 2022**
Bachelor of Engineering in Electronics and Telecommunication Engineering — GPA: 3.6

Indiana University, Luddy School of Informatics, Computing and Engineering **Aug 2024 – May 2026**
Master of Science in Data Science — GPA: 3.6